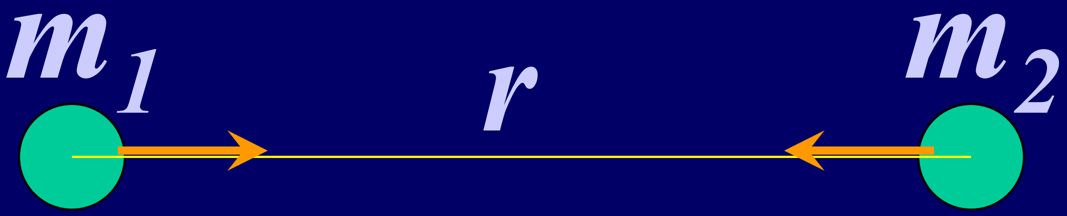
CHAP.14
GRAVITATION



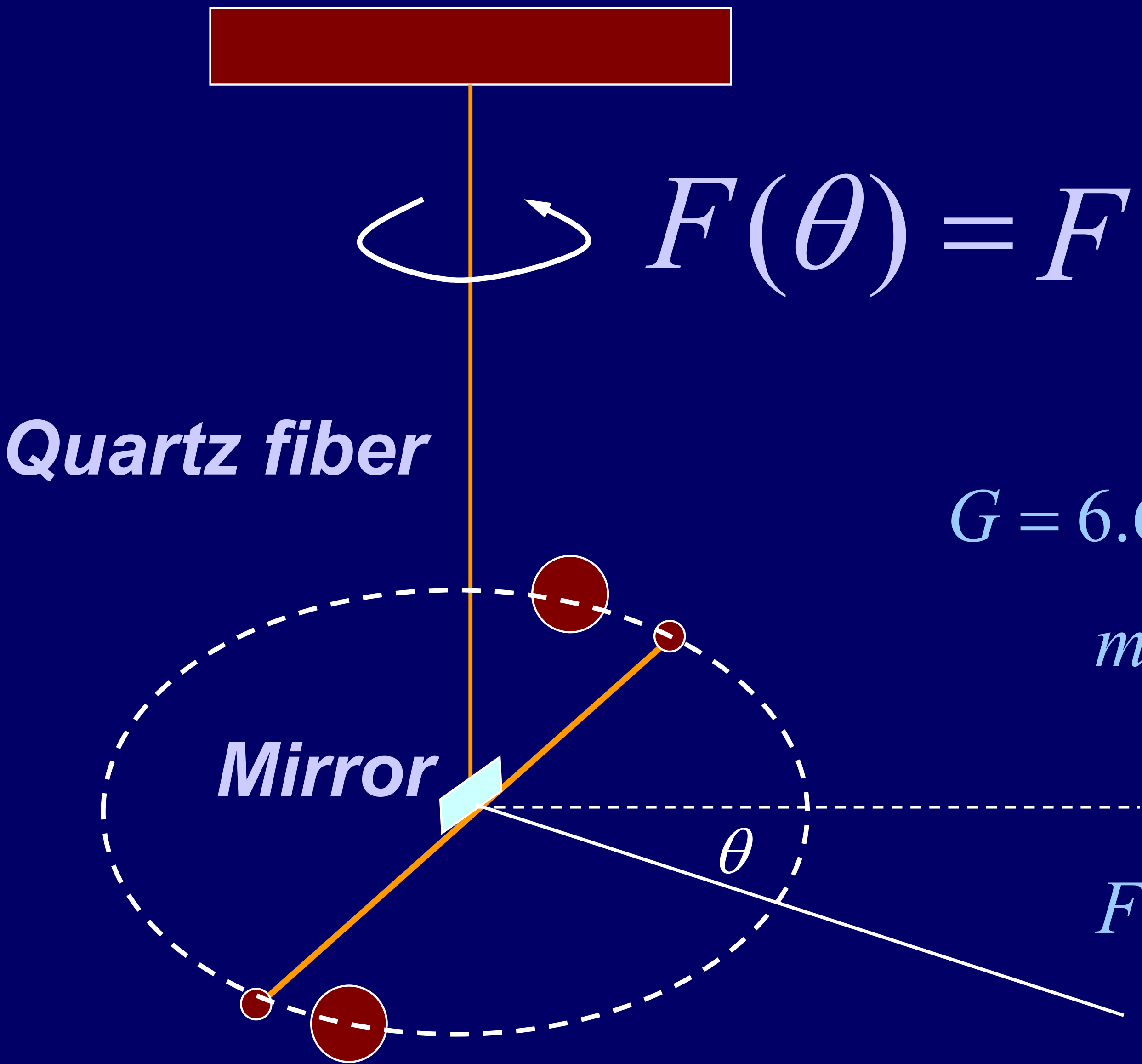
$$F = \frac{Gm_1m_2}{r^2}$$

***G* gravitational constant**

$$\vec{F} = -\frac{Gm_1m_2}{r^2}\hat{u}_r$$

$$\vec{r} = r\hat{u}_r$$

$$\vec{F} = -\frac{Gm_1m_2}{r^3}\vec{r}$$



$$F(\theta) = F = \frac{Gm_1m_2}{r^2}$$

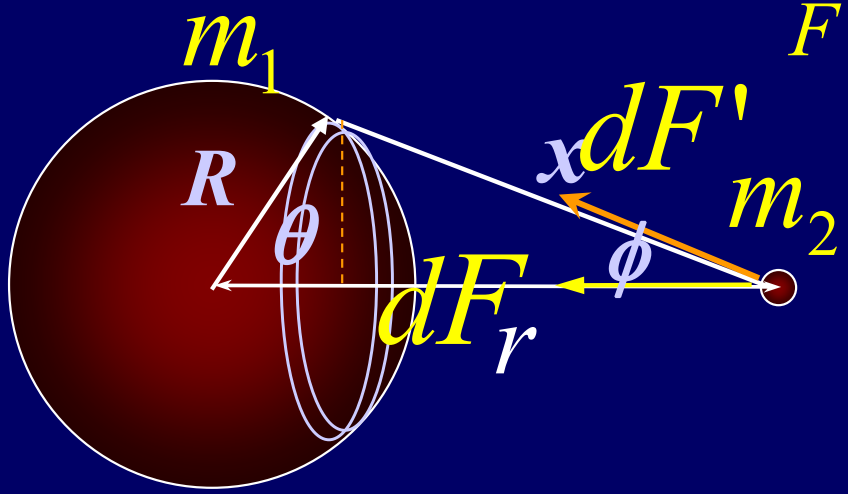
$$G = 6.67 \times 10^{-11}\, N \cdot m^2 / kg^2$$

$$m_1 = m_2 = 100kg$$

$$r = 0.1m$$

$$F = 6.67 \times 10^{-5}\, N$$

Henry Cavendish, 1798



$$F = 2\pi G\sigma R^2 m_2 \int_0^{\pi} \frac{\sin\theta(r - R\cos\theta)d\theta}{(R^2 + r^2 - 2Rr\cos\theta)^{3/2}}$$

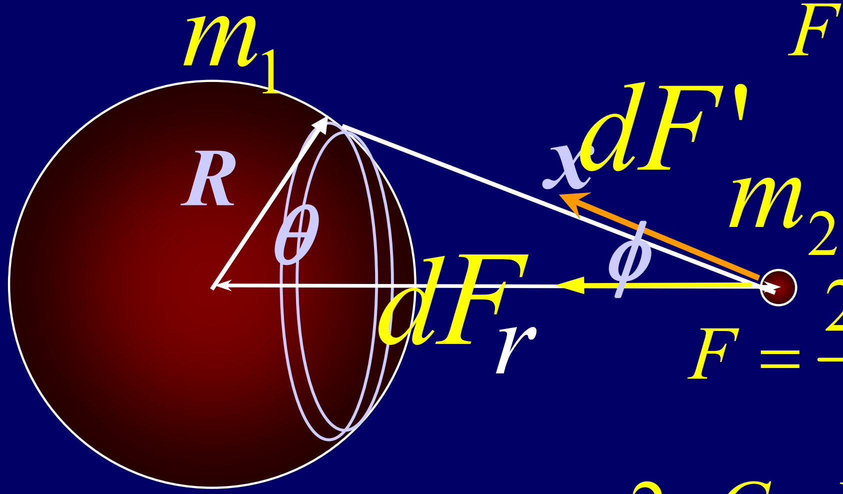
$$dF = \frac{G\sigma ds m_2}{x^2}\cos\phi$$

$$ds = 2\pi R\sin\theta R d\theta \qquad dF = \frac{G\sigma 2\pi\sin\theta R^2 d\theta m_2\cos\phi}{x^2}$$

$$F = \int dF = 2\pi G\sigma R^2 m_2 \int \frac{\sin\theta\cos\phi d\theta}{x^2}$$

$$\cos\phi = \frac{r - R\cos\theta}{x} \qquad x = \sqrt{R^2 + r^2 - 2Rr\cos\theta}$$

$$F = 2\pi G\sigma R^2 m_2 \int_0^{\pi} \frac{\sin\theta (r - R\cos\theta)\, d\theta}{(R^2 + r^2 - 2Rr\cos\theta)^{3/2}}$$



$r > R$

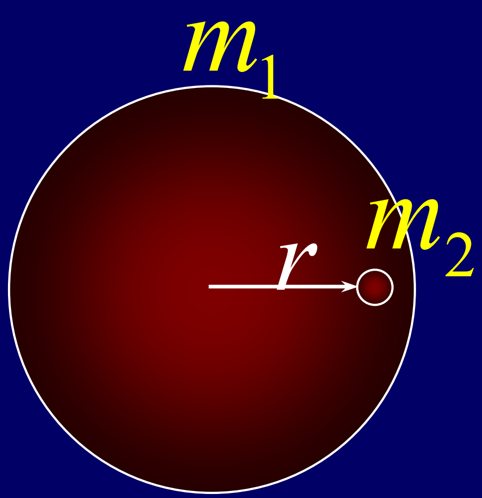
$$F = \frac{2\pi G\sigma R^2 m_2}{2Rr} \int_0^{\pi} \frac{(r - R\cos\theta)\, d(-2Rr\cos\theta)}{(R^2 + r^2 - 2Rr\cos\theta)^{3/2}}$$

$$F = \frac{2\pi G\sigma R^2 m_2}{2Rr \cdot 2r} \int_0^{\pi} \frac{(2r^2 - 2Rr\cos\theta)\, d(-2Rr\cos\theta)}{(R^2 + r^2 - 2Rr\cos\theta)^{3/2}}$$

$$F = \frac{2\pi G\sigma R^2 m_2}{2Rr \cdot 2r} \int_0^{\pi} \frac{(r^2 - R^2 + R^2 + r^2 - 2Rr\cos\theta)\, d(-2Rr\cos\theta)}{(R^2 + r^2 - 2Rr\cos\theta)^{3/2}}$$

$$= \frac{\pi G\sigma R^2 m_2}{2Rr^2} \left(\int_0^{\pi} \frac{(r^2 - R^2)\, d(-2Rr\cos\theta)}{(R^2 + r^2 - 2Rr\cos\theta)^{3/2}} + \int_0^{\pi} \frac{d(-2Rr\cos\theta)}{(R^2 + r^2 - 2Rr\cos\theta)^{1/2}} \right)$$

$$= \frac{\pi G\sigma R^2 m_2}{2Rr^2} (4R + 4R) = \frac{4\pi G\sigma R^2 m_2}{r^2} = \frac{G m_1 m_2}{r^2}$$



$r < R$

$$F = 2\pi G\sigma R^2 m_2 \int_0^{\pi} \frac{\sin\theta(r - R\cos\theta)d\theta}{(R^2 + r^2 - 2Rr\cos\theta)^{3/2}}$$

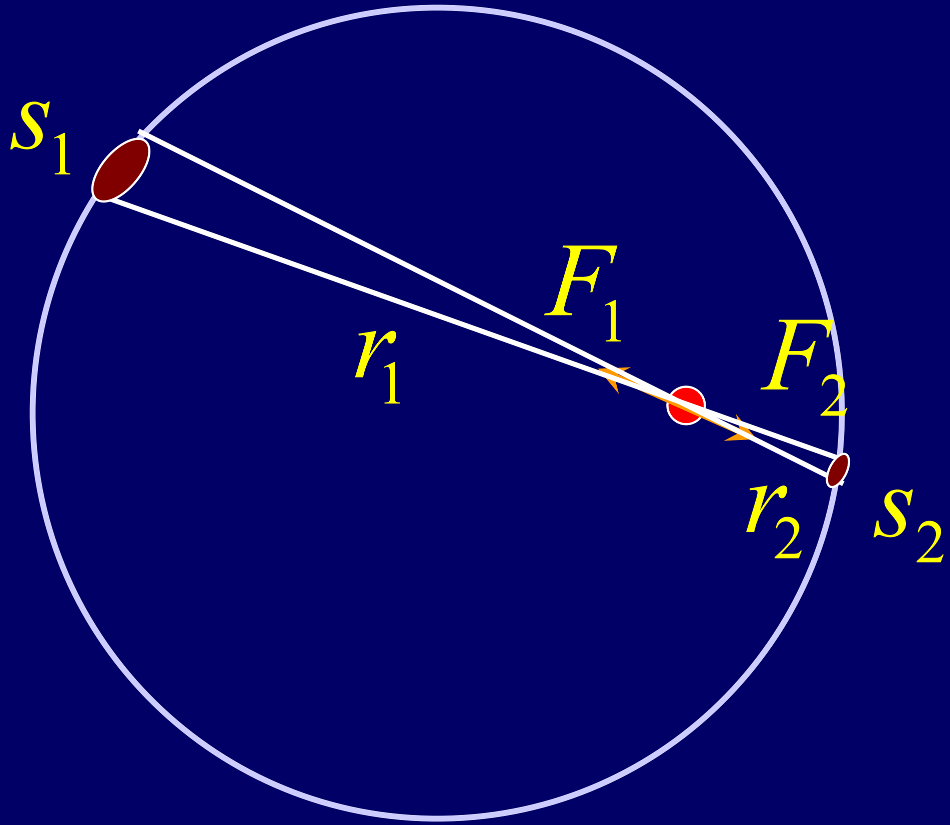
$$F = \frac{2\pi G\sigma R^2 m_2}{2Rr} \int_0^{\pi} \frac{(r - R\cos\theta)d(-2Rr\cos\theta)}{(R^2 + r^2 - 2Rr\cos\theta)^{3/2}}$$

$$F = \frac{2\pi G\sigma R^2 m_2}{2Rr \cdot 2r} \int_0^{\pi} \frac{(2r^2 - 2Rr\cos\theta)d(-2Rr\cos\theta)}{(R^2 + r^2 - 2Rr\cos\theta)^{3/2}}$$

$$F = \frac{2\pi G\sigma R^2 m_2}{2Rr \cdot 2r} \int_0^{\pi} \frac{(r^2 - R^2 + R^2 + r^2 - 2Rr\cos\theta)d(-2Rr\cos\theta)}{(R^2 + r^2 - 2Rr\cos\theta)^{3/2}}$$

$$= \frac{\pi G\sigma R^2 m_2}{2Rr^2}\left(\int_0^{\pi} \frac{(r^2 - R^2)d(-2Rr\cos\theta)}{(R^2 + r^2 - 2Rr\cos\theta)^{3/2}} + \int_0^{\pi} \frac{d(-2Rr\cos\theta)}{(R^2 + r^2 - 2Rr\cos\theta)^{1/2}}\right)$$

$$= \frac{\pi G\sigma R^2 m}{2Rr^2}(-4r + 4r) = 0$$



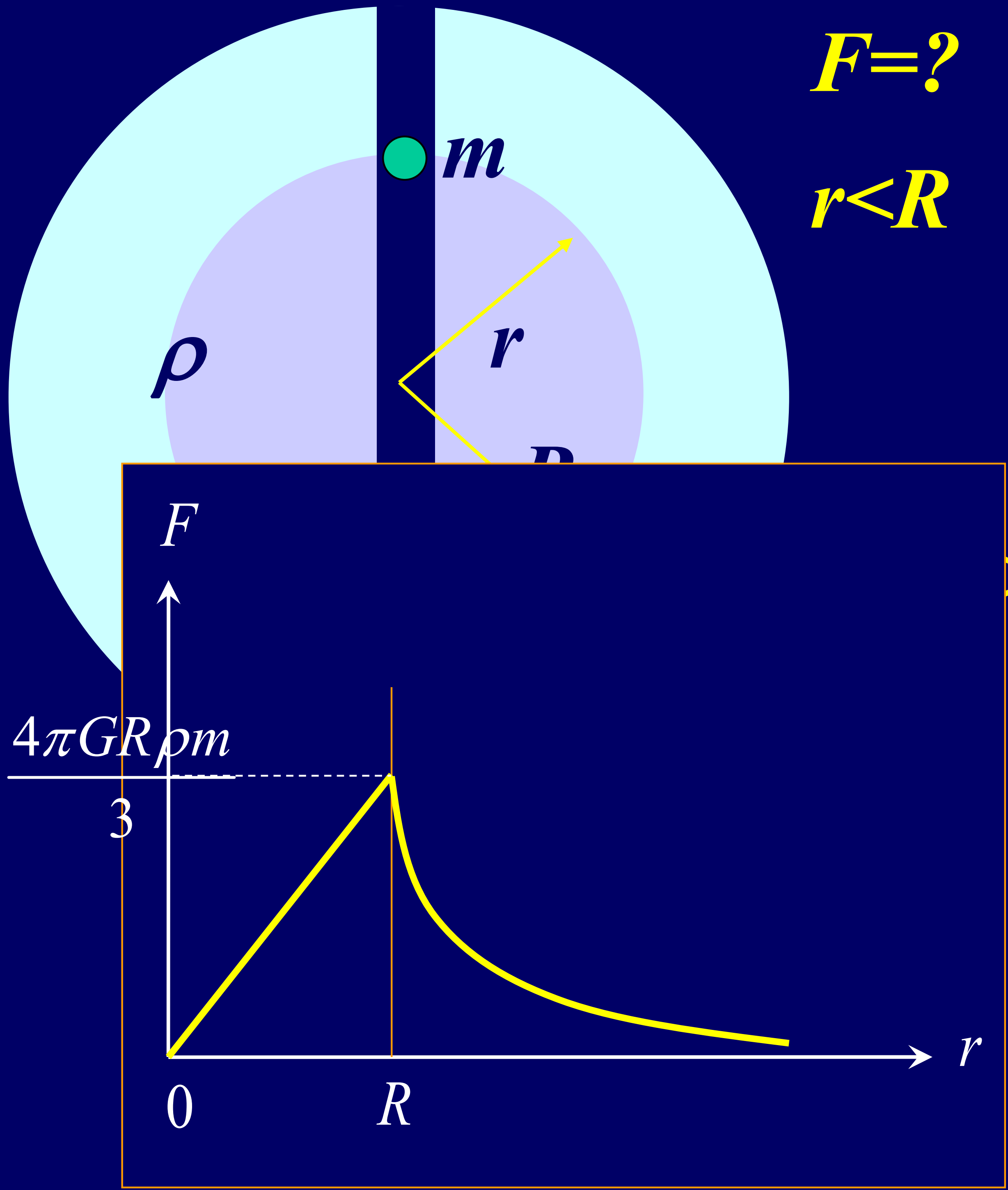
$$|F_1| = \frac{G s_1 \sigma m}{r_1^2}$$

$$|F_2| = \frac{G s_2 \sigma m}{r_2^2}$$

$$\frac{s_2}{r_2^2} = \frac{s_1}{r_1^2}$$

$$\Sigma F = 0$$

$$|F_1| = |F_2|$$



F=?

r<R

$$F = \frac{G\left(\frac{4\pi r^3}{3}\rho\right)m}{r^2}$$

$$F = G\left(\frac{4\pi r}{3}\rho\right)m$$

>R

$$F = \frac{G\left(\frac{4\pi R^3}{3}\rho\right)m}{r^2}$$

$$F = \frac{Gm_1m_2}{r^2}$$

Near earth surface

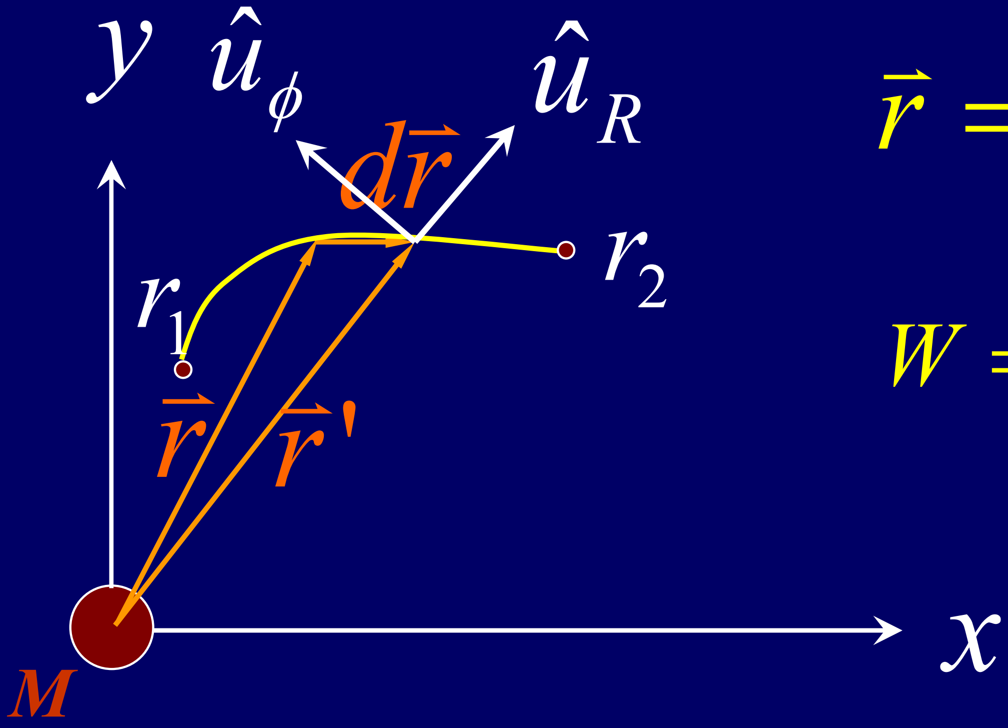
$$F = \frac{G\,M_e\,m}{R_e^{\,2}}$$

$$F = mg$$

$$9.8m/s^2 = g = \frac{G\,M}{R_e^{\,2}}$$

Work

$$W = \int \vec{F} \cdot d\vec{r} \qquad \vec{F} = -\frac{GMm}{r^2}\hat{u}_R$$



$$\vec{r} = r\hat{u}_R \qquad d\vec{r} = dr\hat{u}_R - rd\phi\hat{u}_\phi$$

$$W = \int \vec{F} \cdot d\vec{r} = -\int \frac{GMm}{r^2}\hat{u}_R \cdot dr\hat{u}_R$$

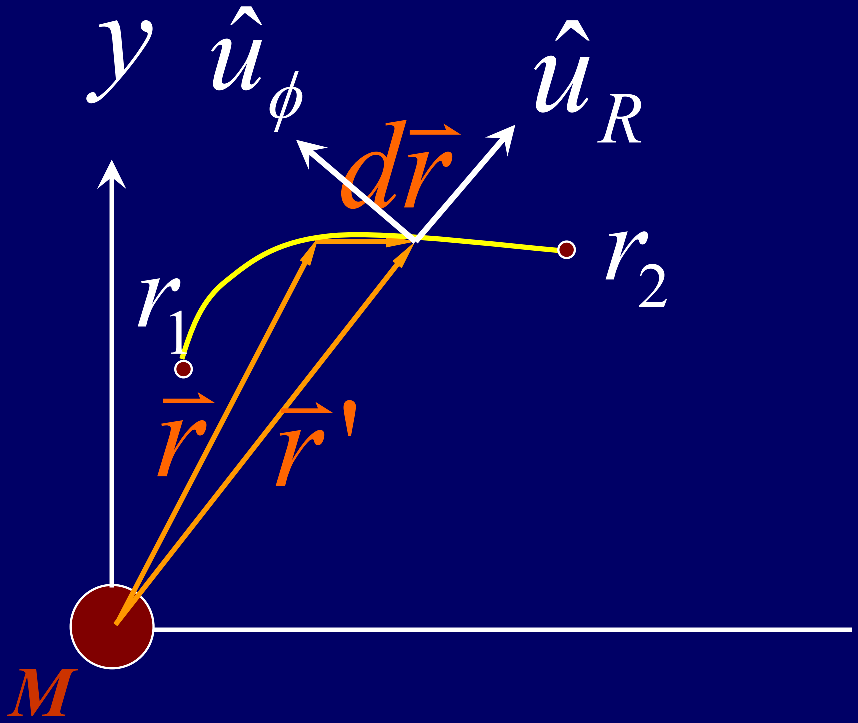
$$W = -\int \frac{GMm}{r^2} dr$$

$$dW = \vec{F} \cdot d\vec{r}$$

$$W = -\int_{r_1}^{r_2} \frac{GMm}{r^2} dr = GMm\left(\frac{1}{r_2} - \frac{1}{r_1}\right)$$

Independent of path

Work



$$dW = \vec{F} \cdot d\vec{r}$$

Gravitational Force is Conservative

Gravitational Potential Energy

$$U(r) = \int_r^{r_0} -\frac{GMm}{r^2}\,dr + U(r_0)$$

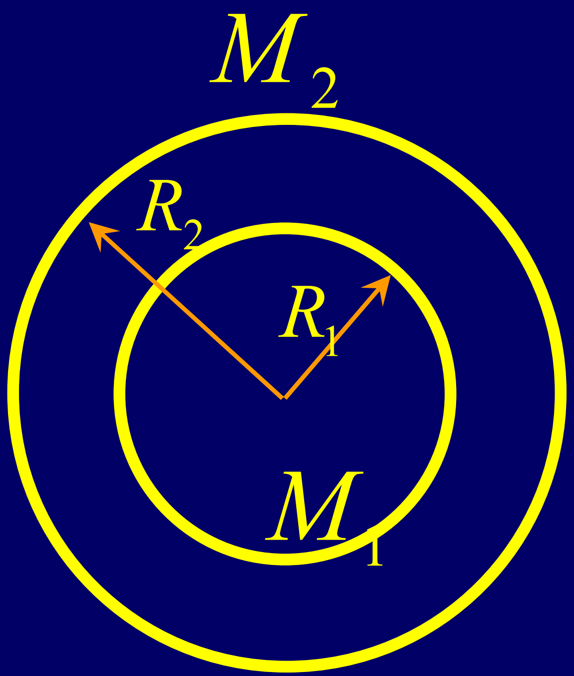
$$U(r) = \int_r^{\infty} -\frac{GMm}{r^2}\,dr$$

$$= GMm\left(\frac{1}{\infty} - \frac{1}{r}\right) = -\frac{GMm}{r}$$

$$W = -\int_{r_1}^{r_2} \frac{GMm}{r^2}\,dr = GMm\left(\frac{1}{r_2} - \frac{1}{r_1}\right)$$

Independent of path

Example



$$a \text{ particle with mass } m$$

$$F = ?$$

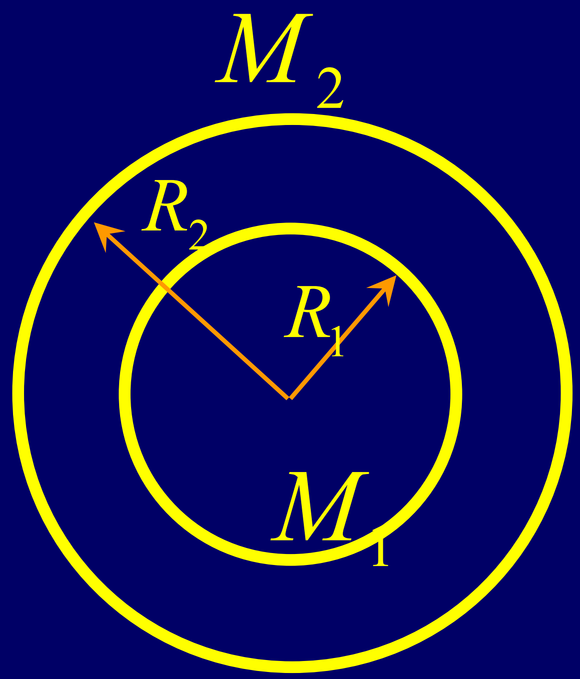
$$r < R_1 \quad F = 0$$

$$R_1 < r < R_2 \quad F = -\frac{GM_1 m}{r^2}$$

$$R_2 < r \quad F = -\frac{G(M_1 + M_2)m}{r^2}$$

a particle with mass m

$$U = ?$$



$$R_2 < r \quad U = -\frac{G(M_1 + M_2)m}{r} = \Sigma U_i$$

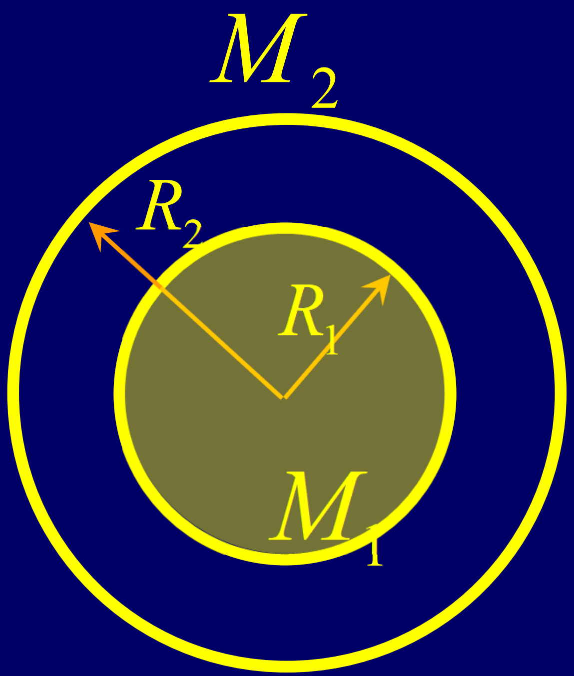
$$R_1 < r < R_2$$

$$U = -\int_{R_2}^{\infty} \frac{G(M_1 + M_2)m}{r^2} dr - \int_{r}^{R_2} \frac{GM_1 m}{r^2} dr$$

$$= -\frac{G(M_1 + M_2)m}{R_2} + \frac{GM_1 m}{R_2} - \frac{GM_1 m}{r}$$

$$= -\frac{GM_2 m}{R_2} - \frac{GM_1 m}{r} = \Sigma U_i$$

$$a \ particle \ with \ mass \ m$$



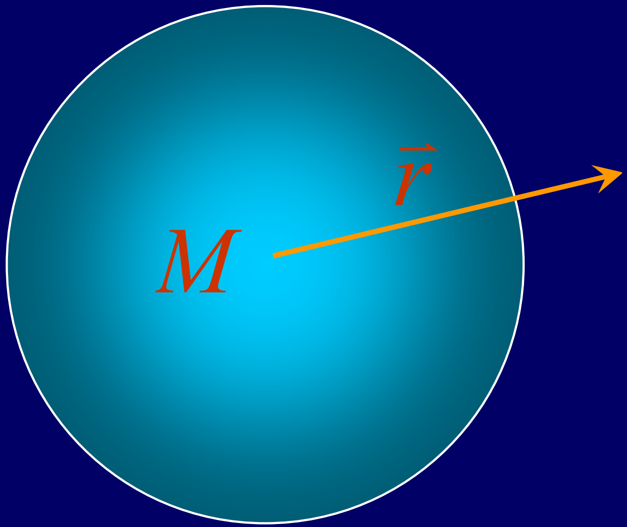
$$U = ?$$

$$r < R_1$$

$$U = -\int_{R_2}^{\infty} \frac{G(M_1 + M_2)m}{r^2} dr - \int_{R_1}^{R_2} \frac{GM_1 m}{r^2} dr$$

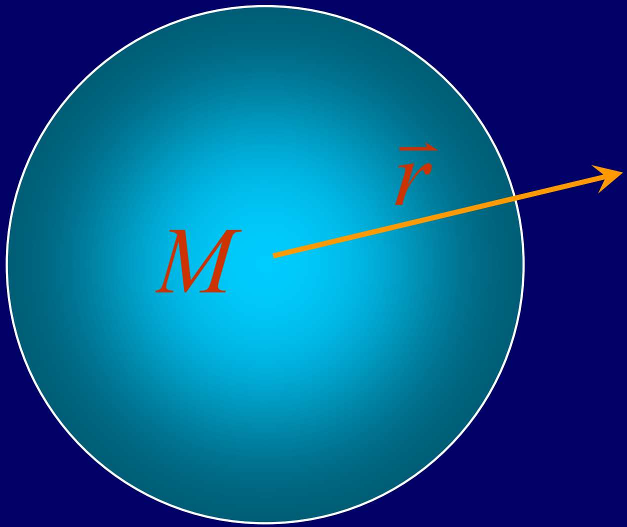
$$= -\frac{G(M_1 + M_2)m}{R_2} + \frac{GM_1 m}{R_2} - \frac{GM_1 m}{R_1}$$

$$= -\frac{GM_2 m}{R_2} - \frac{GM_1 m}{R_1} = \Sigma U_i$$



$$\vec{F} = -\frac{GMm}{r^2}\hat{u}_R = 0$$

$$U = -\frac{GMm}{r} = 0$$



Gravitational Field

$$\vec{E} = \frac{\vec{F}}{m} = -\frac{GM}{r^2}\hat{u}_R$$

Gravitational Potential

$$u = \frac{U}{m} = -\frac{GM}{r}$$

Inertial Mass and Gravitational Mass

Inertial Mass

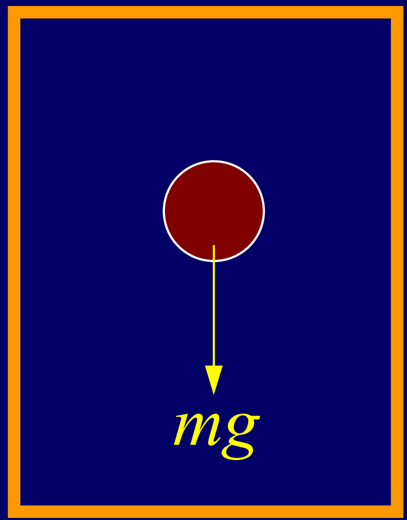
$$\vec{F} = m\vec{a}$$

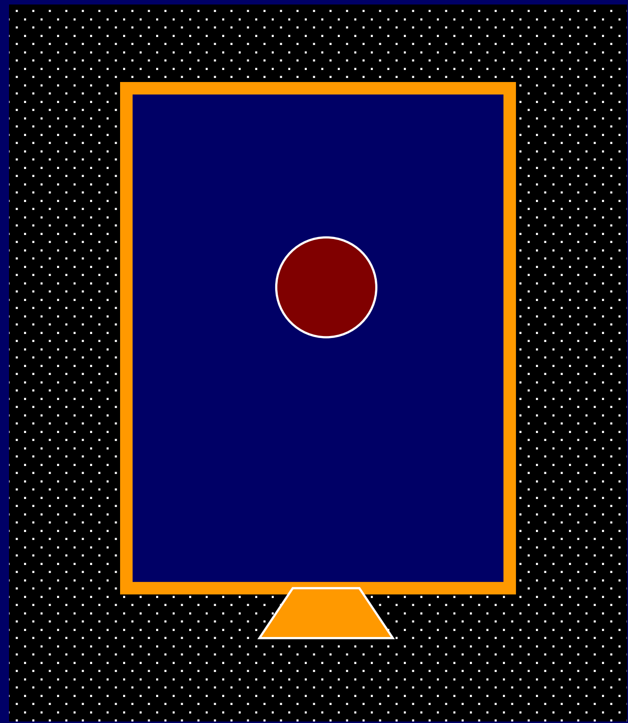
$$\vec{F} = -\frac{GMm}{r^2}\hat{u}_R$$

Gravitational Mass

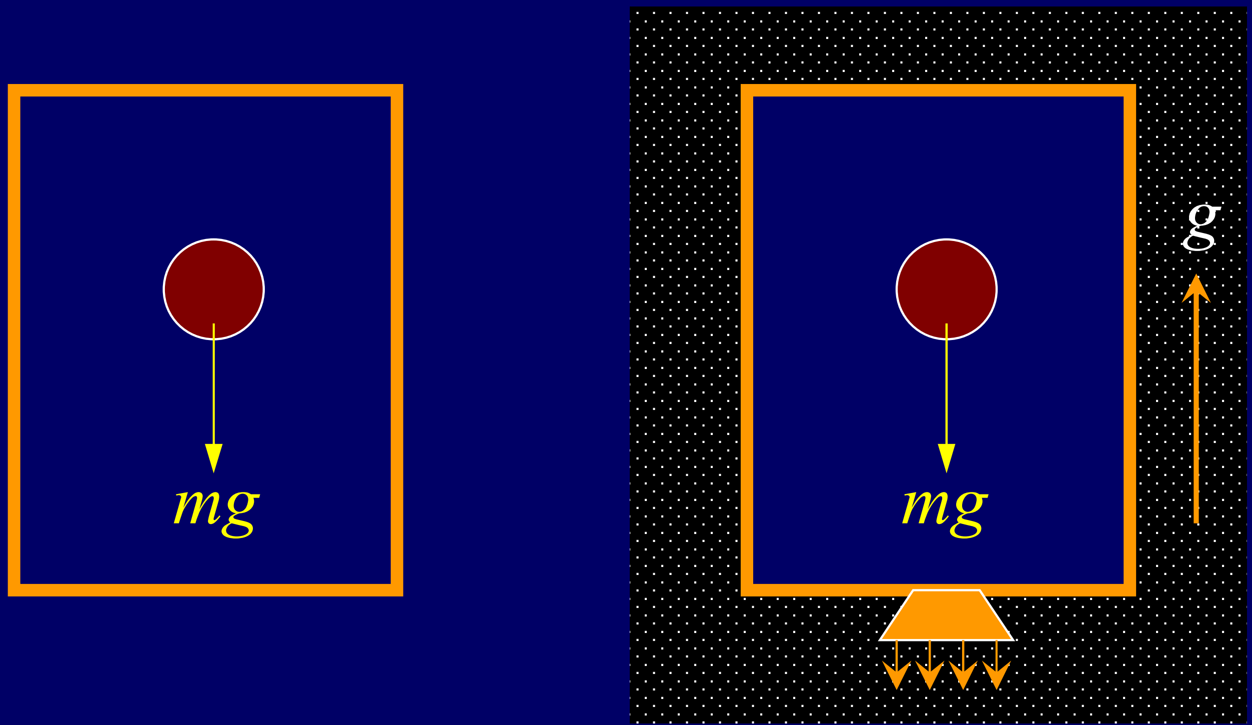
Equivalence

The Principle of Equivalence





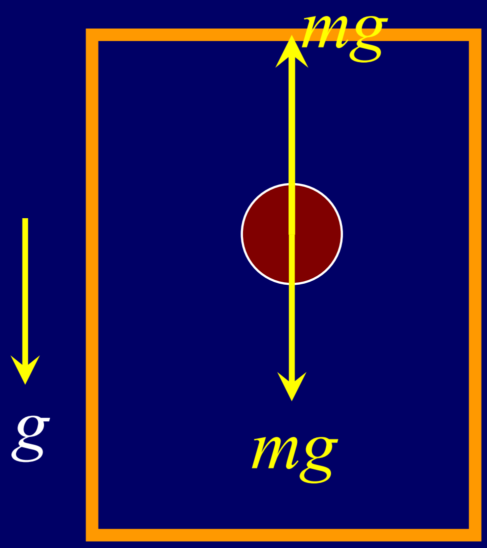
The Principle of Equivalence

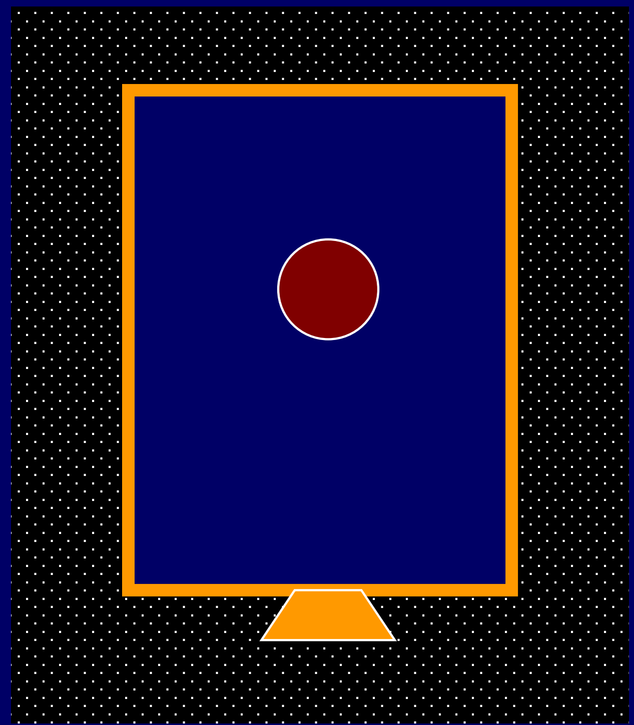


Gravitational Field

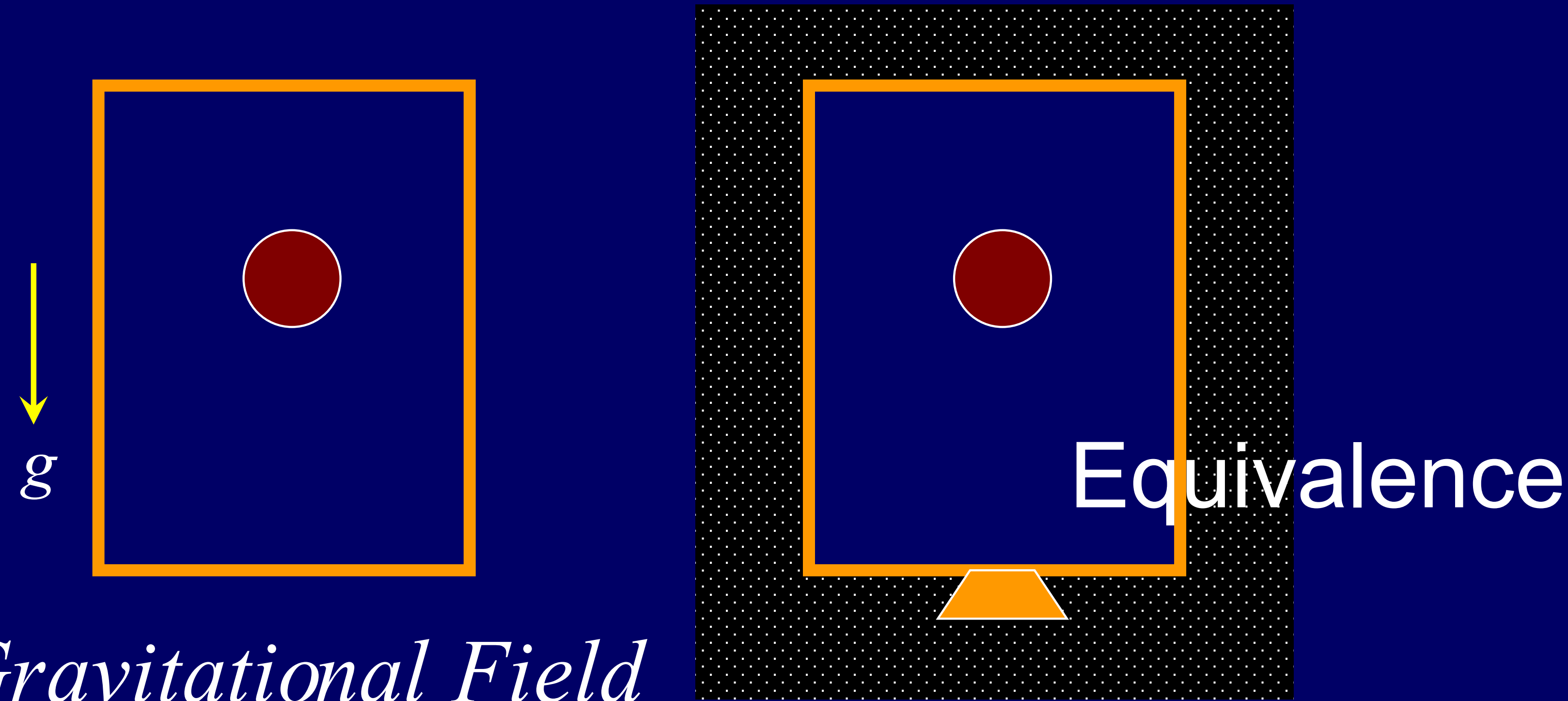
Acceleration Field

The Principle of Equivalence





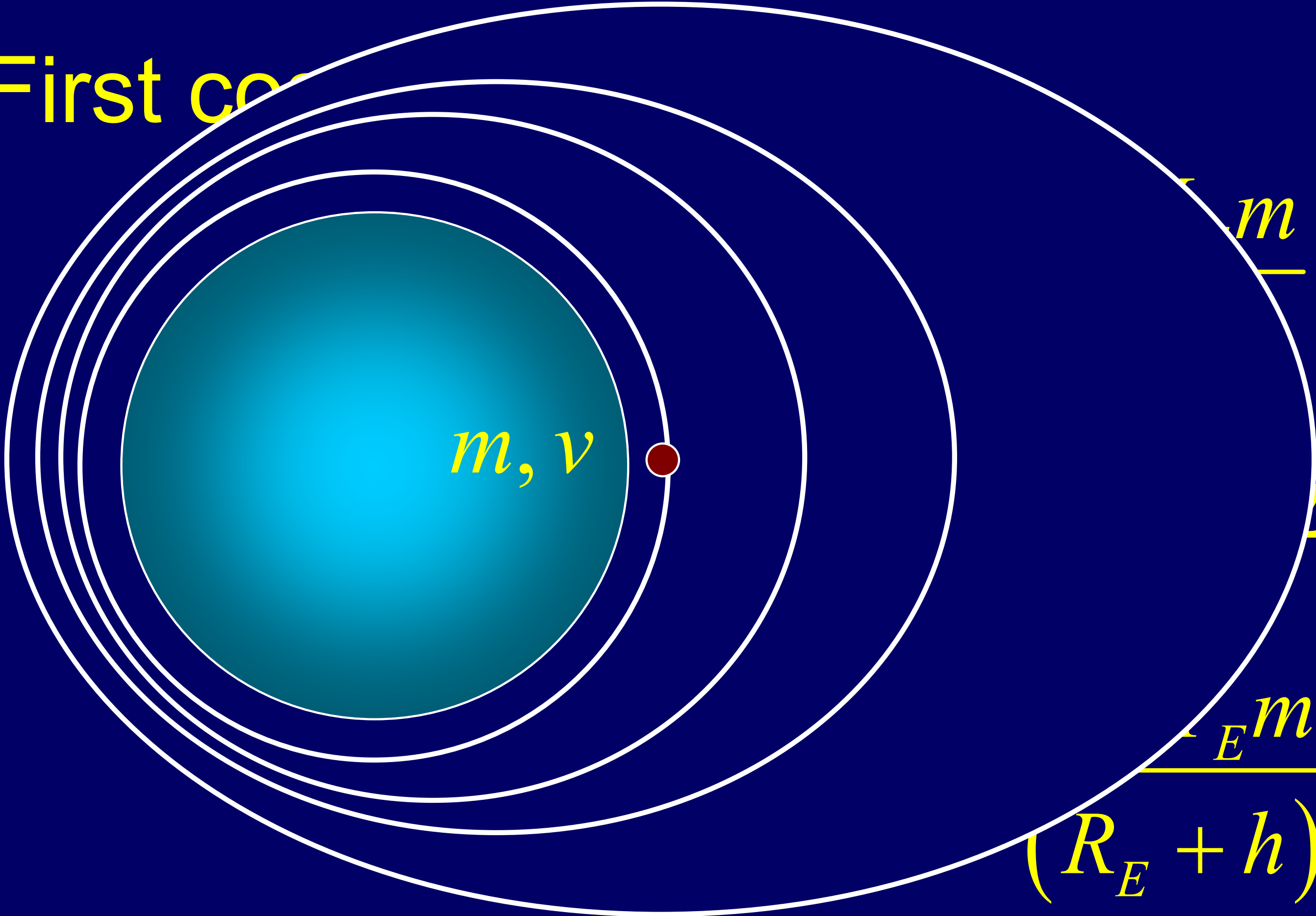
The Principle of Equivalence



Gravitational Field
Acceleration Field

No Field

First co



$$\ldots m = \frac{1}{2}mv^2 - \frac{GM_E m}{R_E + h}$$

$$\ldots 2\frac{GM_E}{R_E + h} + 2\frac{GM_E}{R_E}$$

$$\ldots \frac{\ldots_E m}{(R_E + h)^2} \text{ or } v^2 = \frac{GM_E}{R_E + h}$$

$$v_1 = \sqrt{gR_E\left(2 - \frac{R_E}{R_E + h}\right)} \approx \sqrt{gR_E} = 7.9km/s$$

$$v > v_1 = 7.9km/s$$

Second cosmic velocity or speed for escaping from Earth:

$$E = \frac{1}{2} m {v_2}^2 - \frac{GM_E m}{R_E} = E_{k\infty} + E_{p\infty} = 0$$

$$Escape\ \ Speed \qquad v_2 = \sqrt{\frac{2GM_E}{R_E}} = 11.2 km/s$$

$$Speed\ \ escaping\ \ from\ \ sun \qquad v_3 = \cancel{\sqrt{\frac{2GM_s}{R_s}}}$$

Third cosmic velocity or speed for escaping from Sun:

In Earth's frame $\frac{1}{2}mv_3{}^2 - \frac{GM_E m}{R_E} = \frac{1}{2}mv'^2$ (1)

In Sun's frame $\vec{v}_3' = \vec{v}' + \vec{v}_E \qquad v_3' = v' + v_E$

$$E = \frac{1}{2}mv_3'^2 - \frac{GM_s m}{R_s} = 0 \qquad v_3' = \sqrt{\frac{2GM_s}{R_s}} = 42.2km/s$$

$$v' = v_3' - v_E = \sqrt{\frac{2GM_s}{R_s}} - v_E \approx 12.36\text{km/s} \qquad (2)$$

Speed escaping from sun

$$\frac{M_E v_E{}^2}{R_s} = \frac{GM_E M_s}{R_s{}^2} \text{ or } v_E = \sqrt{\frac{GM_s}{R_s}}$$

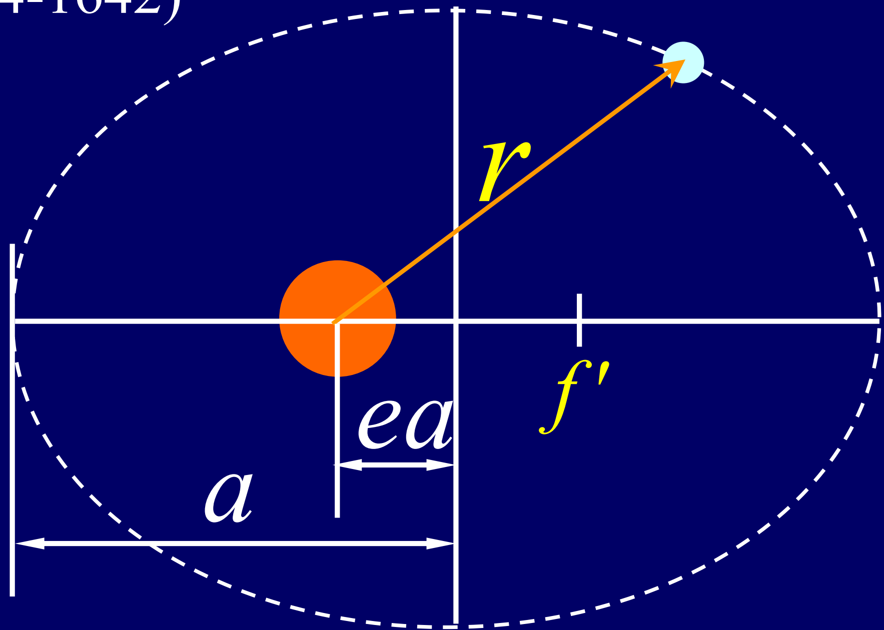
$$v_3 = \sqrt{v'^2 + v_2^2} = 16.7km/s$$

# of Planets and Satellites

Galileo Galilei
(1564-1642)

Kepler's three laws:

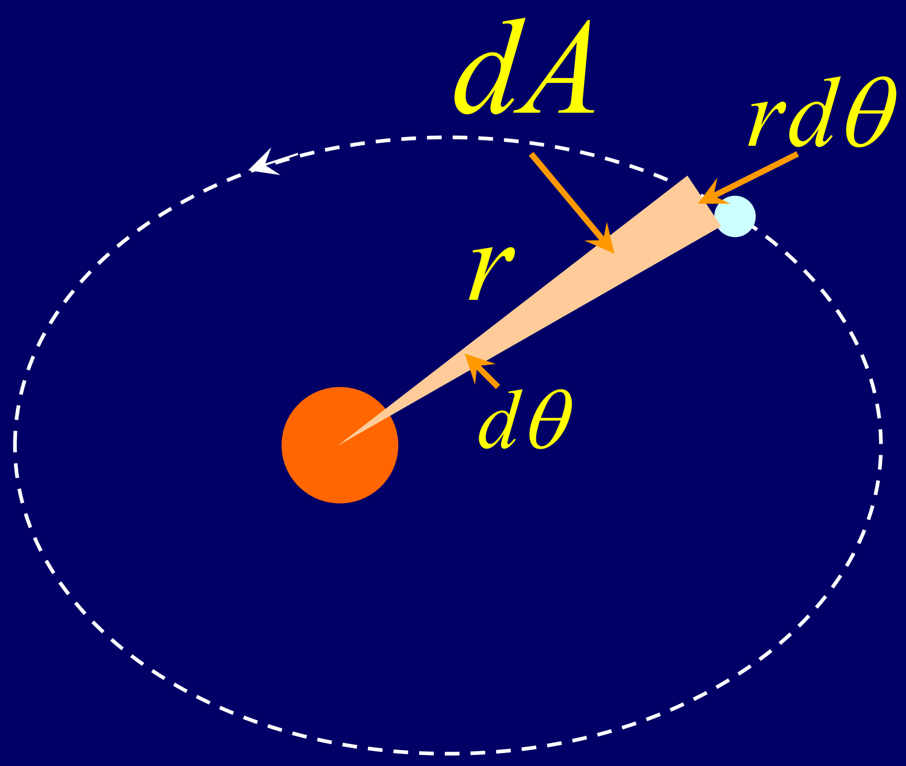
1. **Moves in an elliptical orbit, The sun at one focus**



2. $\frac{dA}{dt} = const$

3. $T^2 \propto r^3$

The Motion of Planets and Satellites



$$\sum \vec{F}_{ext} \neq 0,\ \Delta\vec{p} \neq 0$$

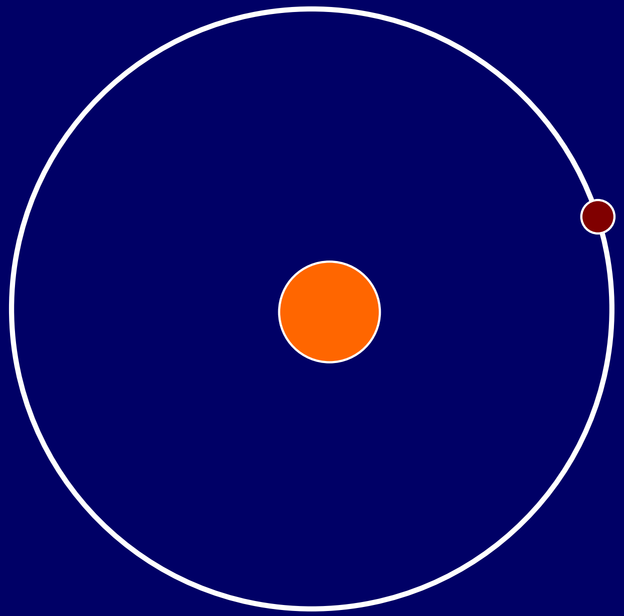
$$\sum \vec{\tau}_{ext} = 0,\ \Delta\vec{L} = 0$$

$$\Delta U + \Delta K = 0$$

$$dA = \frac{1}{2} r \cdot r d\theta$$

$$\frac{dA}{dt} = \frac{1}{2} r^2 \frac{d\theta}{dt} = \frac{L}{2m}$$

$$\boxed{\frac{dA}{dt} = const.}$$



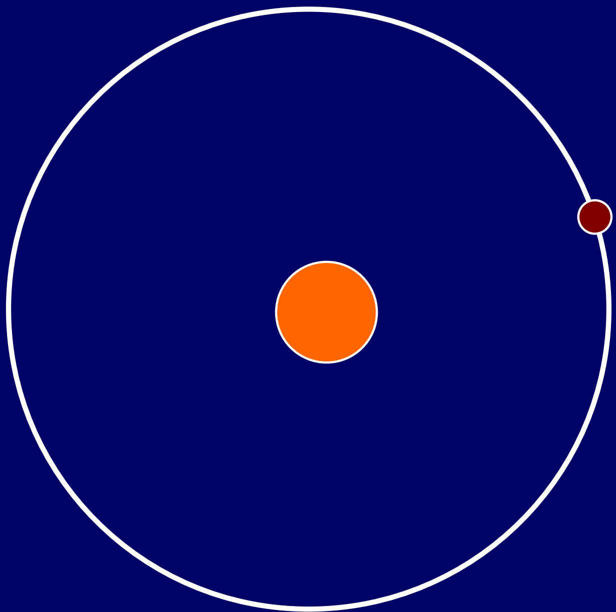
$$\frac{GMm}{r^2} = m\frac{v^2}{r}$$

$$\frac{GMm}{r^2} = m\omega^2 r$$

$$\frac{GMm}{r^3} = m(\frac{2\pi}{T})^2$$

$$\boxed{T^2 = (\frac{4\pi^2}{GM})r^3}$$

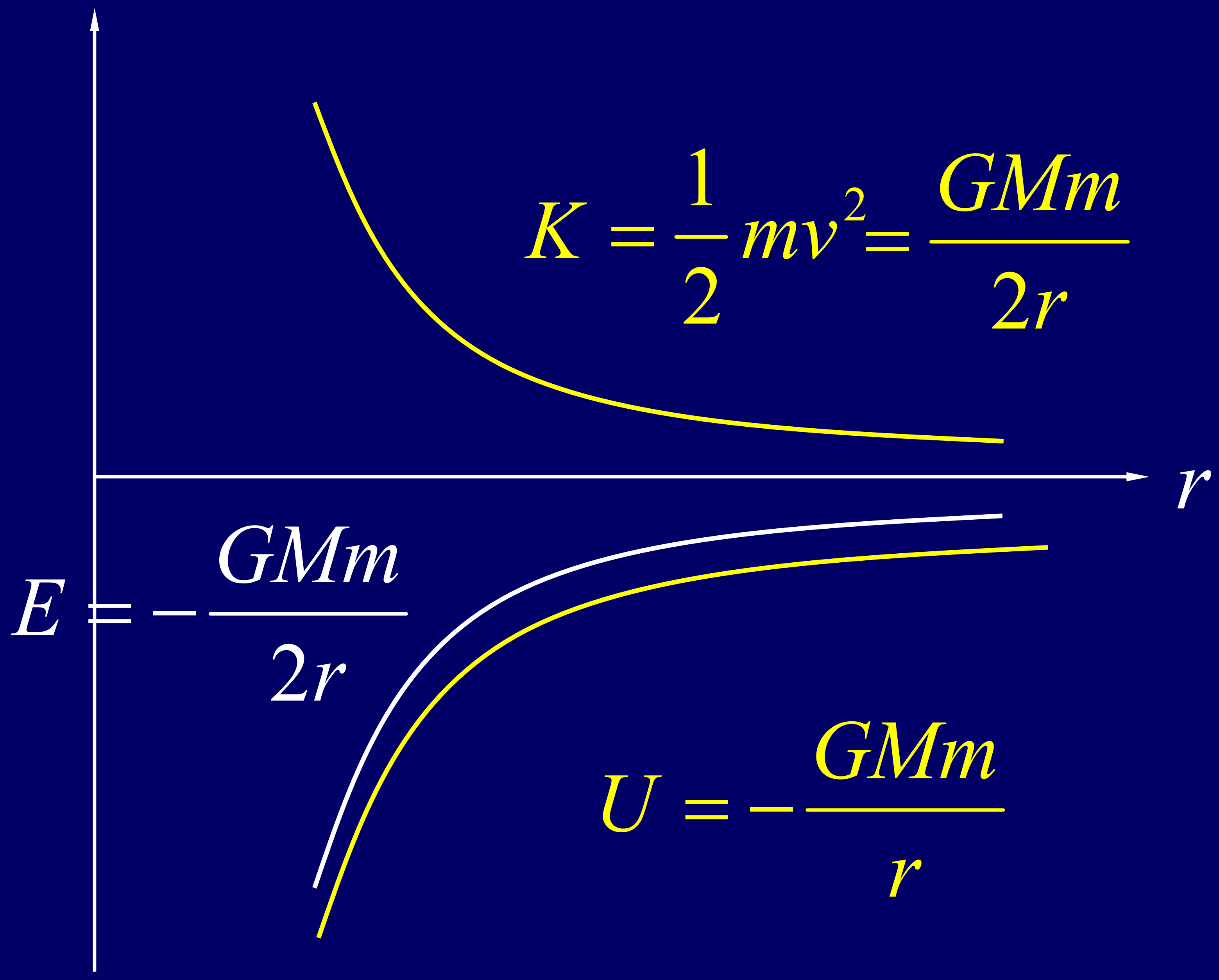
The Energy of Planets and Satellites



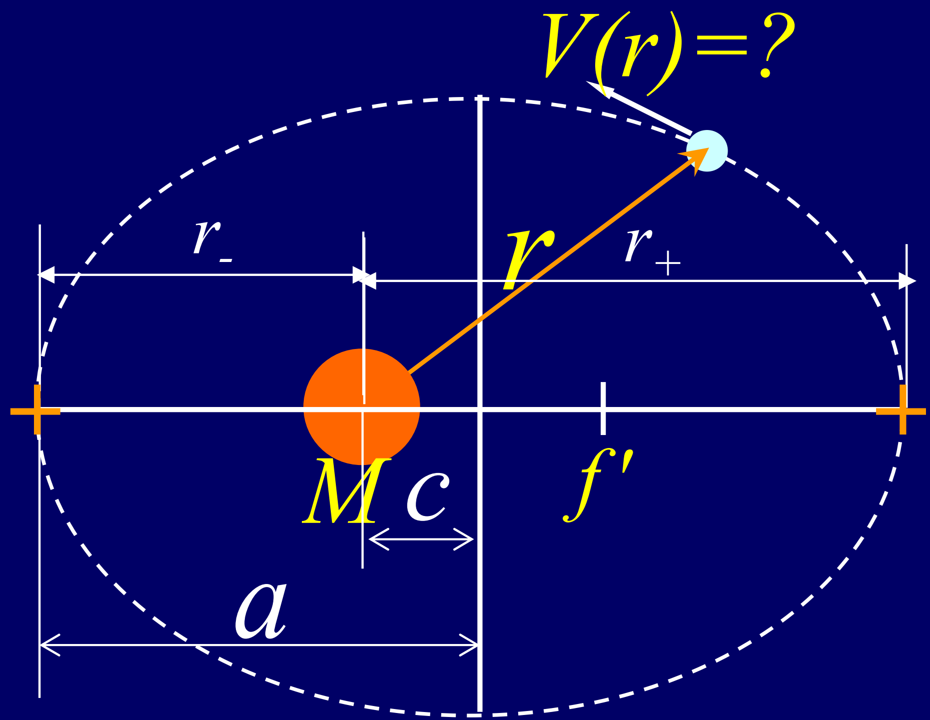
$$U = -\frac{GMm}{r} \qquad K = \frac{1}{2}mv^2$$

$$\frac{mv^2}{r} = \frac{GMm}{r^2} \qquad K = \frac{GMm}{2r}$$

$$E = K + U = \frac{GMm}{2r} - \frac{GMm}{r} = -\frac{GMm}{2r}$$


$K = \frac{1}{2}mv^2 = \frac{GMm}{2r}$
r
$E = -\frac{GMm}{2r}$
$U = -\frac{GMm}{r}$

The Motion of Planets and Satellites



$$E=-\frac{GMm}{r}+\frac{m^2v^2r^2}{2mr^2}$$

$$E=-\frac{GMm}{r}+\frac{L^2}{2mr^2}$$

$$r^2+\frac{GMm}{E}r-\frac{L^2}{2mE}=0$$

$$\frac{1}{2}mv^2-\frac{GMm}{r}=-\frac{GMm}{2a}$$

$$r_\pm=\frac{1}{2}[-\frac{GMm}{E}\pm\sqrt{\frac{G^2M^2m^2}{E^2}+\frac{L^2}{mE}}]$$

$$-\frac{G^2M^2m^3}{L^2}=E \qquad c=0$$

$$c=\frac{r_+-r_-}{2}=\frac{1}{2}\sqrt{\frac{G^2M^2m^2}{E^2}+\frac{L^2}{mE}}$$

CHAP.14

Problems:

P326:11, 13, 14, 18, 22